Words of Concern

Point to be As Allan Bloom has said "Education is the movement from darkness to light". Through this handbook, I have tried to illuminate what might otherwise appear as black boxes to some. In doing so, I have used references from several other authors to synthesize or simplify or elaborate information. This is not possible without omitting details that I deem trivial while dilating the data that I consider relevant to topic. Every effort has been made to avoid errors. In spite of this, some errors might have crept in. Any errors or discrepancies noted maybe brought to my notice which I shall rectify in my next revision.

This handbook is solely for educational purpose and is not for sale. This handbook shall not be reproduced or distributed or used for commercial purposes in any form or by any means.

Thanks,
Sahana K S
Interface College of Computer Applications (ICCA)
Davangere

References

1. The-Morgan-Kaufmann-Series-in-Data-Management-Systems-Jiawei-Han-Micheline-Kamber-Jian-Pei-Data-Mining.-Concepts-and-Techniques-3rd-Edition-Morgan-Kaufmann-2011.
2. https://www.geeksforgeeks.org/data-warehousing/
3. https://aws.amazon.com/what-is/data-science/#:~:text=Data%20science%20is%20the%20study,analyze%20large%20amounts%20of%20data.
4. https://www.blog.trainindata.com/data-science-fundamentals/

Interface College of Computer Applications (ICCA)

Syllabus:

Course Code: DSE-E2

Course Title: Fundamentals Of Data science

Credits - 03

Sem - VI

Hours- 42 Theory

Formative Assessment Marks: 40

Summative Assessment Marks: 60

Duration of SEA/Exam: 02 hrs.

UNIT-1: Data Mining: Introduction, Data Mining Definitions, Knowledge Discovery in Databases (KDD) Vs Data Mining, DBMS Vs Data Mining, DM techniques, Problems, Issues and Challenges in DM, DM applications.

UNIT-2: Data Warehouse: Introduction, Definition, Multidimensional Data Model, Data Cleaning, Data Integration and transformation, Data reduction, Discretization.

UNIT-3: Mining Frequent Patterns: Basic Concept Frequent Item Set Mining Methods-Aprioriand Frequent Pattern Growth (FP Growth) algorithms -Mining Association Rules.

UNIT-4: Classification: Basic Concepts, Issues, Algorithms: Decision Tree Induction. Bayes Classification Methods, Rule-Based Classification, Lazy Learners (or Learning 10 from your Neighbors), k Nearest Neighbor. Prediction - Accuracy- Precision and Recall

UNIT-5: Clustering: Cluster Analysis, Partitioning Methods, Hierarchical Methods. Density-Based Methods, Grid-Based Methods, Evaluation of Clustering

Interface College of Computer Applications (ICCA)

Unit 1 :

Data Mining :

Data Mining is the process of discovering patterns, trends, and useful information from large sets of data. It helps in making smart decisions by turning raw data into meaningful insights.

Data mining is the process of discovering patterns, correlations, and meaningful insights from large volumes of data using various computational techniques. It sits at the intersection of statistics, machine learning, and database systems, transforming raw data into actionable knowledge.

- In other words, we can say that **Data Mining is the process of investigating hidden patterns of information to various perspectives for categorization into useful data,** which is collected and assembled in particular areas such as data warehouses, efficient analysis, data mining algorithms, helping decision making and other data requirement to eventually cost-cutting and generating revenue.
- Data Mining is a process used by organizations to extract specific data from huge databases to solve business problems. It primarily turns raw data into useful information.
- Data mining is one of the most useful techniques that help entrepreneurs, researchers, and individuals to extract valuable information from huge sets of data. Data mining is also called Knowledge Discovery in Database (KDD). The knowledge discovery process includes Data cleaning, Data integration, Data selection, Data transformation, Data mining, Pattern evaluation, and Knowledge presentation.

Interface College of Computer Applications (ICCA)

Core Objectives of Data Mining

Data mining aims to:

1. Predict future trends and behaviors
2. Uncover hidden patterns in complex datasets
3. Support decision-making processes across various domains
4. Transform raw data into valuable business intelligence

Knowledge Discovery in Databases (KDD)

KDD (Knowledge Discovery in Databases) is a process of discovering useful knowledge and insights from large and complex datasets. The KDD process involves a range of techniques and methodologies, including data preprocessing, data transformation, data mining, pattern

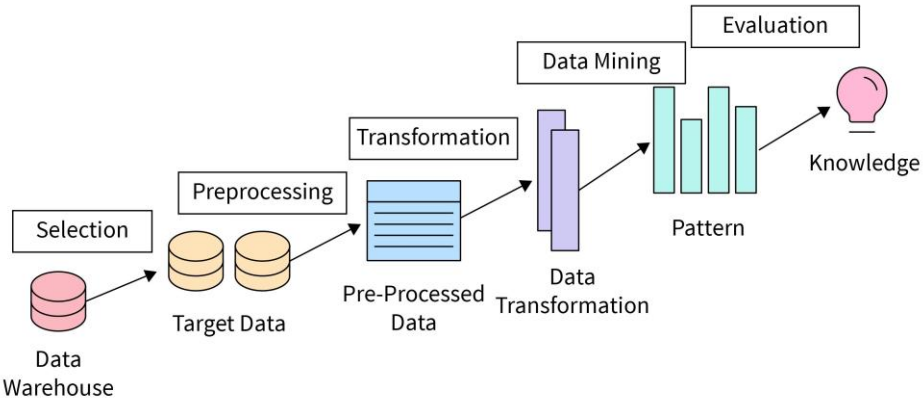
evaluation, and knowledge representation. KDD and data mining are closely related processes, with data mining being a key component and subset of the KDD process.

The KDD process aims to identify hidden patterns, relationships, and trends in data that can be used to make predictions, decisions, and recommendations. KDD is a broad and interdisciplinary field used in various industries, such as finance, healthcare, marketing, e-commerce, etc. KDD is very important for organizations and businesses as it enables them to derive new insights and knowledge from their data, which can be further used to improve decision-making, enhance the customer experience, improve business processes, support strategic planning, optimize operations, and drive business growth.

KDD is widely used in industries like **banking, healthcare, retail, cybersecurity, and social media analysis** to gain insights that help in decision-making.

KDD Process in Data Mining

The KDD process in data mining is a multi-step process that involves various stages to extract useful knowledge from large datasets. The following are the main steps involved in the KDD process -



- **Data Selection** - The first step in the KDD process is identifying and selecting the relevant data for analysis. This involves choosing the relevant data sources, such as databases, data warehouses, and data streams, and determining which data is required for the analysis.
- **Data Preprocessing** - After selecting the data, the next step is data preprocessing. This step involves cleaning the data, removing outliers, and removing missing,

inconsistent, or irrelevant data. This step is critical, as the data quality can significantly impact the accuracy and effectiveness of the analysis.

- **Data Transformation** - Once the data is preprocessed, the next step is to transform it into a format that data mining techniques can analyze. This step involves reducing the data dimensionality, aggregating the data, normalizing it, and discretizing it to prepare it for further analysis.
- **Data Mining** - This is the heart of the KDD process and involves applying various data mining techniques to the transformed data to discover hidden patterns, trends, relationships, and insights. A few of the most common data mining techniques include clustering, classification, association rule mining, and anomaly detection.
- **Pattern Evaluation** - After the data mining, the next step is to evaluate the discovered patterns to determine their usefulness and relevance. This involves assessing the quality of the patterns, evaluating their significance, and selecting the most promising patterns for further analysis.
- **Knowledge Representation** - This step involves representing the knowledge extracted from the data in a way humans can easily understand and use. This can be done through visualizations, reports, or other forms of communication that provide meaningful insights into the data.
- **Deployment -** The final step in the KDD process is to deploy the knowledge and insights gained from the data mining process to practical applications. This involves integrating the knowledge into decision-making processes or other applications to improve organizational efficiency and effectiveness.

Advantages of KDD in Data Mining

Interface College of Computer Applications (ICCA)

KDD in data mining is a powerful approach for extracting useful knowledge and insights from large datasets. It is very important for organizations as it has a lot of advantages. Some of the advantages of KDD in data mining are -

- **Helps in Decision Making** - KDD can help make informed and data-driven decisions by discovering hidden patterns, trends, and relationships in data that might not be immediately apparent.
- **Improves Business Performance** - KDD can help organizations improve their business performance by identifying areas for improvement, optimizing processes, and reducing costs.

- **Saves Time and Resources** - KDD can help save time and resources by automating the data analysis process and identifying the most relevant and significant information or knowledge.
- **Increases Efficiency** - KDD can help organizations streamline their processes, optimize their resources, and increase their overall efficiency.
- **Enhances Customer Experience** - KDD can help organizations improve customer experience by understanding customer behavior, preferences, and requirements and giving personalized products and services.
- **Fraud Detection** - KDD can help detect fraud and identify fraudulent behavior by analyzing patterns in data and identifying anomalies or unusual behavior.
- **Enables Predictive Modeling -** KDD can enable organizations to develop predictive models that can forecast future trends and behaviors, providing a competitive advantage in the market.

Disadvantages of KDD in Data Mining

- While KDD (Knowledge Discovery in Databases) is a powerful approach to extracting useful knowledge and insights from large datasets, there are also some potential disadvantages to consider -
- **Requires High-Quality Data -** KDD relies on high-quality data to generate accurate and meaningful insights. If the data is incomplete, inconsistent, or of poor quality, it can lead to inaccurate, misleading results and flawed conclusions.
- **Complexity -** KDD is a complex and time-consuming process that requires specialized skills and knowledge to perform effectively. The complexity can also make interpreting and communicating the results challenging to non-experts.
- **Privacy and Compliance Concerns -** KDD can raise ethical concerns related to privacy, compliance, bias, and discrimination. For example, data mining techniques can extract sensitive information about individuals without their consent or reinforce existing biases or stereotypes.
- **High Cost -** KDD can be expensive, and require specialized software, hardware, and skilled professionals to perform the analysis. The cost can be prohibitive for smaller organizations or those with limited resources.

Interface College of Computer Applications (ICCA)

Factors	KDD process	Data Mining
Definition	It is a comprehensive process that includes multiple steps for	A subset of KDD that focuses primarily on finding

	extracting useful knowledge and insights from large datasets	patterns and relationships in data
Steps involved	It includes steps such as data collection, cleaning, integration, selection, transformation, data mining, interpretation, and evaluation	It includes steps such as data preprocessing, modeling, and analysis
Focus	Emphasizes the importance of domain expertise in interpreting and validating results	Focuses on the use of computational algorithms to analyze data
Techniques used	Data selection, cleaning, transformation, data mining, pattern evaluation, interpretation, knowledge representation, and data visualization	Association rules mining, clustering, regression, classification, and dimensionality reduction.
Outputs	Knowledge bases, such as rules or models that help organizations make informed decisions	A set of patterns, relationships, predictions, or insights to support decision-making or business understanding

ICCA
Interface College of Computer Applications (ICCA)

Challenges and Future Trends in KDD

- Scalability issues in handling large datasets
- Dealing with noisy, incomplete, and inconsistent data
- Incorporating unstructured data sources (text, images, video)
- Addressing privacy and security concerns in KDD
- Emerging trends: automated machine learning (AutoML), explainable AI (XAI), federated learning
- The role of cloud computing and big data technologies in KDD
- Ethical considerations in KDD: bias detection and mitigation, fairness, transparency. Example: implementing fairness-aware machine learning algorithms.

- **DBMS and Data Mining**

- **Attribute DBMS Data Mining**

Definition	A software system that manages databases and provides access to stored data.	The process of discovering patterns and extracting useful information from large datasets.
Purpose	Store, manage, and retrieve data efficiently.	Extract knowledge and insights from data.
Data Storage	Structured data stored in tables with predefined schemas.	Structured, semi-structured, and unstructured data stored in various formats.
Data Manipulation	Supports CRUD operations (Create, Read, Update, Delete) on data.	Performs complex operations like clustering, classification, and regression on data.
Query Language	SQL (Structured Query Language) is commonly used.	SQL, as well as specialized languages like R and Python, are used.
Focus	Efficient data storage, retrieval, and transaction processing.	Pattern discovery, knowledge extraction, and predictive modeling.
Usage	Used in day-to-day operations of organizations to manage and process data.	Used in research, business intelligence, and decision-making processes.
Examples	Oracle, MySQL, SQL Server.	Weka, RapidMiner, Apache Spark.

Interface College of Computer Applications (ICCA)

DBMS and Data Mining

Database Management Systems (DBMS) and Data Mining are two essential components in the field of data management and analysis. While DBMS focuses on efficiently storing, retrieving, and managing structured data, Data Mining aims to extract meaningful patterns and

insights from large datasets. Both play crucial roles in modern data-driven decision-making processes. In this article, we will explore and compare the attributes of DBMS and Data Mining, highlighting their unique features and applications.

DBMS: Efficient Data Storage and Retrieval

A DBMS is a software system designed to manage databases, providing an interface for users to interact with the stored data. One of the primary attributes of DBMS is its ability to efficiently store and retrieve data. It organizes data into tables, rows, and columns, allowing for structured and organized storage. DBMS ensures data integrity by enforcing data constraints, such as primary keys and foreign keys, preventing inconsistencies and redundancies.

Furthermore, DBMS provides powerful query languages, such as SQL (Structured Query Language), enabling users to retrieve specific data based on various conditions. These queries can involve complex joins, aggregations, and filtering operations, allowing for flexible and precise data retrieval. DBMS also supports indexing techniques, such as B-trees, to optimize query performance by reducing the number of disk accesses required.

Another crucial attribute of DBMS is its support for concurrent access and transaction management. Multiple users can simultaneously access and modify the data without conflicts, thanks to concurrency control mechanisms like locking and timestamp-based protocols. DBMS ensures the atomicity, consistency, isolation, and durability (ACID) properties of transactions, guaranteeing data integrity and reliability.

In summary, DBMS excels in efficient data storage, retrieval, query processing, and transaction management, making it an indispensable tool for managing structured data in various domains.

Interface College of Computer Applications (ICCA)

Data Mining: Extracting Insights from Big Data

Data Mining, on the other hand, focuses on extracting valuable patterns, trends, and insights from large datasets. It employs various techniques, including statistical analysis, machine learning, and pattern recognition, to discover hidden relationships and knowledge within the data.

One of the primary attributes of Data Mining is its ability to handle vast amounts of data. With the exponential growth of data in recent years, traditional analysis methods often fall short. Data Mining algorithms can efficiently process and analyze massive datasets, enabling organizations to uncover valuable information that can drive decision-making processes.

Data Mining algorithms can be broadly categorized into two types: supervised and unsupervised learning. Supervised learning algorithms aim to predict or classify data based on labeled examples. They learn from historical data to make predictions on new, unseen data. Unsupervised learning algorithms, on the other hand, explore the data without predefined labels, identifying patterns and clusters that may not be immediately apparent.

Another crucial attribute of Data Mining is its ability to handle diverse data types. While DBMS primarily deals with structured data, Data Mining can handle structured, semi-structured, and unstructured data. This flexibility allows organizations to analyze a wide range of data sources, including text documents, images, videos, social media feeds, and more.

Data Mining also plays a vital role in various applications, such as customer segmentation, fraud detection, market analysis, recommendation systems, and predictive maintenance. By uncovering hidden patterns and insights, organizations can make data-driven decisions, optimize processes, and gain a competitive edge.

Comparison: DBMS vs. Data Mining

While DBMS and Data Mining serve different purposes, they are interconnected and complement each other in the data management and analysis pipeline. Let's compare their attributes:

1. Data Handling

DBMS focuses on structured data, providing efficient storage, retrieval, and management capabilities. It ensures data integrity and supports complex querying operations. On the other hand, Data Mining can handle diverse data types, including structured, semi-structured, and unstructured data. It employs advanced algorithms to extract patterns and insights from large datasets.

Interface College of Computer Applications (ICCA)

2. Purpose

DBMS is primarily used for data storage, retrieval, and transaction management. It serves as the backbone of applications that require structured data management, such as e-commerce platforms, banking systems, and inventory management systems. Data Mining, on the other hand, focuses on knowledge discovery and decision support. It helps organizations uncover hidden patterns, make predictions, and gain insights to drive business strategies.

3. Techniques

DBMS relies on indexing, query optimization, and transaction management techniques to ensure efficient data access and integrity. It provides powerful query languages like SQL for data retrieval. Data Mining, on the other hand, employs statistical analysis, machine learning, and pattern recognition techniques to discover patterns, clusters, and relationships within the data. It utilizes algorithms like decision trees, neural networks, and association rules to extract valuable insights.

4. Applications

DBMS finds applications in various domains that require structured data management, such as finance, healthcare, logistics, and human resources. It ensures data consistency, reliability, and security. Data Mining, on the other hand, is widely used in marketing, finance, healthcare, fraud detection, and recommendation systems. It helps organizations gain a competitive advantage by uncovering hidden patterns and trends.

5. Data Volume

DBMS is efficient in handling moderate-sized datasets, ensuring fast data retrieval and transaction processing. However, it may face challenges when dealing with big data, which requires distributed systems and parallel processing. Data Mining, on the other hand, is specifically designed to handle large datasets. It leverages distributed computing and parallel processing techniques to analyze massive amounts of data efficiently.

ADVANTAGES OF DATA MINING

Interface College of Computer Applications (ICCA)

Data mining is a powerful tool used across various industries to extract meaningful patterns and knowledge from large datasets. Here are some of the key advantages:

1. **Informed Decision-Making**: Data mining helps organizations make better decisions by uncovering hidden patterns and relationships in data. This can lead to more accurate forecasting, improved strategic planning, and effective resource allocation.

2. **Improved Customer Relationships**: By analysing customer data, companies can better understand customer preferences and behaviours. This enables personalized marketing, targeted promotions, and improved customer service, which can enhance customer satisfaction and loyalty.

3. **Enhanced Efficiency**: Data mining can identify inefficiencies and areas for improvement within processes and operations. For example, it can help streamline supply chain management, optimize production schedules, and reduce operational costs.

4. **Competitive Advantage**: Organizations can gain a competitive edge by leveraging data mining to uncover trends and insights that their competitors might not be aware of. This can lead to innovative products, services, and business strategies.

5. **Risk Management**: Data mining techniques can help in identifying potential risks and fraud. For instance, financial institutions use data mining to detect unusual transaction patterns that could indicate fraudulent activity, allowing for timely intervention.

6. **Predictive Analytics**: By analysing historical data, data mining can provide predictive insights into future trends and behaviours. This is valuable for anticipating market changes, customer needs, and other business dynamics.

7. **Product and Service Improvement**: Data mining can reveal insights into product performance and customer feedback, leading to improvements in product design, quality, and service offerings.

8. **Enhanced Data Quality**: Data mining can identify inconsistencies, errors, and missing values in datasets, leading to better data quality and more reliable analysis.

9. **Market Segmentation**: Businesses can use data mining to segment their markets more effectively. This allows for more precise targeting of different customer groups and tailoring of marketing strategies to meet the needs of each segment.
10. **Trend Analysis**: Data mining can help identify and analyze emerging trends in various fields, such as consumer behavior, market conditions, and technological advancements. This information can be crucial for adapting business strategies and staying ahead of the curve.

Interface College of Computer Applications (ICCA)

ISSUES OR CHALLENGES DATA MINING

Data mining, while powerful, also comes with a set of challenges and issues that can impact its effectiveness and ethical implications. Here are some of the key challenges:

1. **Data Quality**: Poor-quality data—such as incomplete, inaccurate, or inconsistent data—can lead to misleading or incorrect findings. Ensuring data cleanliness and accuracy is crucial for reliable results.

2. **Data Privacy**: Mining sensitive or personal data raises significant privacy concerns. There is a risk of unauthorized access and misuse of personal information, which can lead to ethical and legal issues.

3. **Data Security**: Protecting data from breaches, cyber-attacks, and other security threats is a major concern. Ensuring robust security measures is essential to safeguard sensitive information.

4. **Complexity and Scalability**: As datasets grow larger and more complex, the computational resources and time required for data mining increase. Handling large-scale data efficiently requires advanced technologies and infrastructure.

5. **Bias and Fairness**: Data mining algorithms can perpetuate or even exacerbate biases present in the data. If the data reflects historical biases or imbalances, the results can lead to unfair or discriminatory outcomes.

6. **Interpretability**: Some data mining techniques, such as deep learning models, can be complex and hard to interpret. Understanding how these models make decisions is important for trust and accountability.

7. **Ethical Considerations**: The ethical use of data mining involves ensuring that data is used responsibly and that decisions based on data mining do not harm individuals or communities. This includes considerations around consent and transparency.

8. **Legal and Regulatory Compliance**: Adhering to data protection regulations, such as GDPR in Europe or CCPA in California, is essential. Non-compliance can result in legal penalties and damage to reputation.
9. **Integration with Existing Systems**: Integrating data mining insights into existing business processes and systems can be challenging. It requires careful planning and coordination to ensure that the insights are actionable and effectively utilized.
10. **Overfitting**: Data mining models can sometimes be too closely aligned with the training data, resulting in overfitting. This means the model performs well on the training data but poorly on new, unseen data.

Interface College of Computer Applications (ICCA)

11. **Skill Requirements**: Effective data mining requires expertise in statistics, machine learning, and domain knowledge. Finding and retaining skilled professionals can be a challenge.
12. **Changing Data Patterns**: Data patterns can change over time due to evolving market conditions, behaviors, or other factors. Models need to be updated regularly to reflect these changes and remain accurate.

Applications of Data Mining

1. Business Intelligence and Marketing

Businesses use data mining extensively to understand customer behavior and optimize operations:

Customer Segmentation: Companies analyze purchasing patterns to group customers with similar behaviors. For example, an online retailer might discover distinct shopping patterns among different age groups, allowing them to tailor marketing campaigns specifically for each segment.

Market Basket Analysis: This technique identifies products frequently purchased together. A grocery store might discover that customers who buy diapers often buy beer as well, leading to strategic product placement that increases sales of both items.

Recommendation Systems: These systems suggest products or content based on past behavior. When Amazon recommends products you might like or Netflix suggests shows based on your viewing history, they're using data mining techniques that analyze patterns across millions of users.

Interface College of Computer Applications (ICCA)

2. Healthcare and Medicine

The healthcare industry employs data mining to improve patient outcomes and operational efficiency:

Disease Prediction and Diagnosis: By analyzing patient records, genetic information, and environmental factors, healthcare providers can identify individuals at risk for specific diseases. For instance, patterns in blood test results combined with lifestyle factors might reveal patients at high risk for diabetes before symptoms appear.

Treatment Optimization: Mining patient records helps determine the most effective treatments for particular conditions. Analysis of treatment outcomes across thousands of similar cases can reveal which approaches work best for specific patient profiles.

Fraud Detection: Healthcare providers and insurers use data mining to identify suspicious billing patterns. Unusual combinations of treatments or billing irregularities that match known fraud patterns can be flagged for investigation.

3. Financial Services

Financial institutions rely heavily on data mining for risk management and service improvement:

Credit Scoring: Banks analyze income, payment history, and demographic data to predict loan repayment probability. This helps determine whether to approve loans and what interest rates to charge.

Fraud Detection: Financial institutions use data mining to identify unusual transaction patterns that may indicate fraud. For example, a credit card purchase that deviates significantly from a customer's usual spending patterns might trigger a security alert.

Stock Market Analysis: Traders use data mining to analyze market trends and predict stock movements. Algorithms can process news articles, financial reports, and historical price data to identify potential investment opportunities.

Risk Management: Insurance companies analyze claim history, demographic information, and external data sources to accurately price policies and manage risk exposure.

Interface College of Computer Applications (ICCA)

4. Scientific Research

Data mining has revolutionized how scientists conduct research:

Climate Science: Scientists mine weather patterns and historical climate data to improve prediction models and understand climate change. This involves processing terabytes of satellite imagery, temperature readings, and atmospheric measurements.

5. Government and Public Sector

Government agencies use data mining for various public service improvements:

Tax Compliance: Tax authorities analyze filing patterns to identify potential evasion. Unusual deductions or income reporting that differs significantly from similar taxpayers might trigger an audit.

Crime Analysis: Law enforcement agencies mine crime reports to identify patterns and predict high-risk areas. This "predictive policing" helps allocate officers more effectively to prevent crime.

Social Service Optimization: Government agencies mine program participation data to improve service delivery and prevent benefit fraud. This helps ensure resources reach those most in need.

6 Education

Educational institutions use data mining to improve learning outcomes:

Student Performance Prediction: Schools analyse academic and demographic data to identify students at risk of falling behind. A university might discover that specific patterns of class attendance, assignment completion, and early quiz scores strongly predict final course outcomes.

Curriculum Optimization: Data mining helps identify which teaching methods and materials are most effective. An online learning platform might discover that short video lessons followed by interactive exercises lead to better knowledge retention than longer lectures.

Personalized Learning: Adaptive learning systems use data mining to tailor educational content to individual students. By analyzing how students interact with different types of problems, these systems can present material that matches each student's learning style and current knowledge level.

Interface College of Computer Applications (ICCA)

Prediction
Goal: Predict continuous numeric values.

How it works: Uses existing data to predict unknown or future values.

Techniques:

Linear Regression

Multiple Regression

Data Mining Techniques

Several key techniques power these applications:

Classification

- Classification is one of the most important techniques in data mining and machine learning. It involves predicting which category or class a data instance belongs to, based on previously observed data whose categories are already known.

- classification is a supervised learning process that assigns items to predefined categories. The process works in two main phases:
- **Learning Phase (Training)**: The algorithm learns from a training dataset where each instance has a known label or class. During this phase, the algorithm identifies patterns or relationships between the input features and the target classes.
- **Classification Phase (Testing/Prediction)**: Once trained, the model applies the learned patterns to classify new, previously unseen instances.

Key Elements in Classification

Several components are crucial to the classification process:

- **Training Data**: This is the labeled dataset used to train the model. Each instance consists of:
- Feature vectors (independent variables or predictors)
- Class labels (dependent variable or target)

  Features: These are the attributes or characteristics of the data that the algorithm uses to make predictions. For example, when classifying emails as spam or not spam, features might include the presence of certain words, sender information, and formatting characteristics.

  Target Variable: This is what we're trying to predict - the class or category assignment. It's typically categorical (nominal) but can also be ordinal.

  Classification Model: The mathematical representation that captures patterns in the training data and can be used to predict classes for new instances.

  Decision Boundary: This is the border that a classification algorithm establishes to separate different classes in the feature space.

Interface College of Computer Applications (ICCA)

Clustering

Clustering is an unsupervised machine learning technique used in data mining to group similar data points together. Unlike classification, clustering does not require labeled data. Instead, it identifies hidden patterns and structures in data by organizing similar objects into groups (or clusters).

The Essence of Clustering

At its core, clustering aims to partition data into groups (clusters) where:

- Objects within the same cluster are highly similar to each other

- Objects in different clusters are distinctly dissimilar from each other

This similarity is typically measured using distance metrics in a multi-dimensional space where each dimension represents a different attribute or feature of the data.

Clustering algorithms operate in a feature space where each data point is represented as a vector. The clustering task can be formulated mathematically as finding a partition of the dataset that optimizes certain criteria.

For a dataset with n points, a clustering algorithm partitions it into k clusters C_1, C_2, ..., C_k such that:

- Each point belongs to exactly one cluster (in hard clustering)
- The union of all clusters equals the entire dataset
- The clusters are non-empty

The optimization typically involves minimizing the within-cluster variation or maximizing the between-cluster separation.

Clustering groups similar items without predefined categories. A streaming service might use clustering to group viewers with similar tastes, even if they don't have explicit genre preferences.

Association Rule Learning

This technique discovers relationships between variables. Supermarkets use it to identify products frequently purchased together, informing inventory management and store layout decisions.

Interface College of Computer Applications (ICCA)

Regression Analysis

Regression predicts continuous values rather than categories. Real estate companies use regression to estimate property values based on features like location, size, and age.

Anomaly Detection

This technique identifies unusual data points that differ significantly from the majority. Credit card companies use anomaly detection to flag transactions that deviate from a customer's normal spending patterns.

Neural Networks and Deep Learning

These advanced techniques model complex patterns through interconnected layers of processing nodes. Image recognition, natural language processing, and autonomous driving rely heavily on these approaches.

Ethical Considerations and Challenges

With great power comes great responsibility:

Privacy Concerns: Data mining often involves personal information, raising significant privacy issues. Companies must balance insight generation with respect for individual privacy rights.

Bias and Fairness: Algorithms can perpetuate existing biases in training data. For example, a hiring algorithm trained on historical hiring decisions might discriminate against certain demographic groups if past hiring practices were biased.

Transparency: Many advanced data mining techniques produce "black box" models whose decision-making processes are difficult to explain. This creates challenges for accountability, especially in high-stakes applications like healthcare or criminal justice.

Data Quality Issues: Poor quality data leads to unreliable results. Missing values, inconsistent formats, and measurement errors can significantly impact analysis outcomes.

Scalability Challenges: As data volumes grow, processing and analyzing them becomes increasingly complex. Organizations must continually upgrade their infrastructure and techniques to handle expanding datasets.

Interface College of Computer Applications (ICCA)

. Association Rule Learning
Goal: Discover relationships between variables in large datasets.

Used in: Market Basket Analysis

Popular Algorithm:

Apriori Algorithm

FP-Growth Algorithm

Regression
Goal: Understand the relationship between dependent and independent variables.

Used in: Prediction and forecasting

Types:

Linear Regression

Anomaly Detection (Outlier Detection)
Goal: Identify rare or unusual data records that differ from the majority.

Used in: Fraud detection, network intrusion detection.

Example: Detecting a fraudulent transaction in banking.

Unit 2

Data Warehouse:

A data warehouse is a centralized repository designed for storing, managing, and analyzing large volumes of structured data from multiple sources. It enables businesses to perform complex queries, generate reports, and conduct data analysis efficiently.

A data warehouse, or enterprise data warehouse (EDW), is a system that aggregates data from different sources into a single, central, consistent data store to support data analysis, data mining, artificial intelligence (AI), and machine learning.

Data warehousing provides architectures and tools for business executives to systematically organize, understand, and use their data to make strategic decisions.

Data warehouse systems are valuable tools in today's competitive, fast-evolving world. In the last several years, many firms have spent millions of dollars in building enterprise-wide data warehouses. Many people feel that with competition mounting in every industry, data warehousing is the latest must-have marketing weapon, a way to retain customers by learning more about their needs.

Data warehouses have been defined in many ways, making it difficult to formulate a rigorous definition. Loosely speaking, **a data warehouse refers to a database that is maintained separately from an organization's operational databases. Data warehouse systems allow for the integration of a variety of application systems. They support information iprocessing by providing a solid platform of consolidated historical data for analysis.**

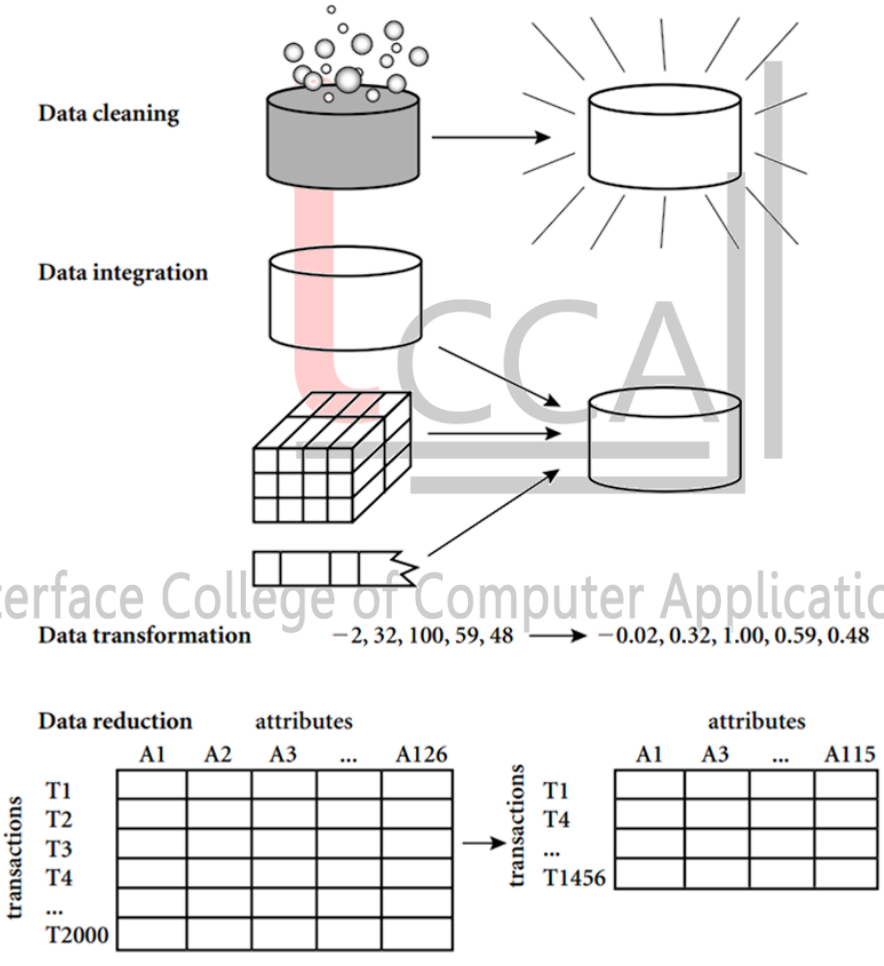
Interface College of Computer Applications (ICCA)

DATA PREPROCESSING

Data preprocessing is an important step in the data mining process. **It refers to cleaning, transforming, and integrating data to prepare it for analysis. The goal of data preprocessing is to improve the data quality and make it more suitable for the specific data mining task.**

Incomplete, noisy, and inconsistent data are commonplace properties of large real-world databases and data warehouses. Incomplete data can occur for several reasons.

- Attributes of interest may not always be available, such as customer information for sales transaction data.
- Some data may not be included simply because it was not considered important at the time of entry. Relevant data may not be recorded due to a misunderstanding, or because of equipment malfunctions.
- Data that were inconsistent with other recorded data may have been deleted.
- The data collection instruments used may be faulty. There may have been human or computer errors occurring at data entry.
- Errors in data transmission can also occur.
- Incorrect data may also result from inconsistencies in naming conventions or data codes used, or inconsistent formats for input fields, such as date. Duplicate tuples also require data cleaning



ICCA
Interface College of Computer Applications (ICCA)

Fig: Forms of data preprocessing

DATA CLEANING

- Data cleaning help us remove inaccurate, incomplete and incorrect data from the dataset.
- Clean the data by filling in missing values, smoothing noisy data, identifying or removing outliers, and resolving inconsistencies.
- Dirty data can cause confusion for the mining procedure, resulting in unreliable output.
- **Some techniques used in data cleaning are**
    - **Handling missing values**
        - This type of scenario occurs when some data is missing.
    - **Noisy Data**
        - Noisy data are the data that cannot be interpreted by machine and It can be generated due to faulty data collection, data entry errors.

1. Missing value

Imagine that you need to analyze AllElectronics sales and customer data. You note that many tuples have no recorded value for several attributes, such as customer income. How can you go about filling in the missing values for this attribute?

Let's look at the following methods:

- **Ignore the tuple:** This is usually done when the class label is missing (assuming the mining task involves classification). This method is not very effective unless the tuple contains several attributes with missing values. It is especially poor when the percentage of missing values per attribute varies considerably.
- **Fill in the missing value manually:** In general, this approach is time-consuming and may not be feasible given a large data set with many missing values.
- **Use a global constant to fill in the missing value:** Replace all missing attribute values with the same constant, such as a label like "Unknown" or $-\infty$. If missing values are replaced by, say, "Unknown," then the mining program may mistakenly think that they form an interesting concept since they all have a value in common—that of "Unknown." Hence, although this method is simple, it is not foolproof.
- **Use the attribute mean to fill in the missing value:** For example, suppose that the average income of AllElectronics customers is $56,000. Use this value to replace the missing value for income.
- **Use the attribute mean for all samples belonging to the same class as the given tuple:** For example, if classifying customers according to credit risk, replace the missing

Interface College of Computer Applications (ICCA)

value with the average income value for customers in the same credit risk category as that of the given tuple.

- **Use the most probable value to fill in the missing value:** This may be determined with regression, inference-based tools using a Bayesian formalism, or decision tree induction. For example, using the other customer attributes in your data set, you may construct a decision tree to predict the missing values for income

Noisy Data

Noisy data is a meaningless data that can't be interpreted by machines.It can be generated due to faulty data collection, data entry errors etc. It can be handled in following ways :

1. **BinningMethod:**

   This method works on sorted data in order to smooth it. The whole data is divided into segments of equal size and then various methods are performed to complete the task. Each segmented is handled separately. One can replace all data in a segment by its mean or boundary values can be used to complete the task.

   - **Smoothing by bin mean:** each value in a bin is replaced by the mean value of the bin.

     For example, the mean of the values 4, 8, and 15 in Bin 1 is 9. Therefore, each original value in this bin is replaced by the value 9.
   - **Smoothing by bin median:** each bin value is replaced by the bin median.
   - **Smoothing by bin boundry:** the minimum and maximum values in a given bin are identified as the bin boundaries. Each bin value is then replaced by the closest boundary value. In general, the larger the width, the greater the effect of the smoothing.

ICCA
r Applications (ICCA)

Sorted data for *price* (in dollars): 4, 8, 15, 21, 21, 24, 25, 28, 34

Partition into (equal-frequency) bins:
Bin 1: 4, 8, 15
Bin 2: 21, 21, 24
Bin 3: 25, 28, 34

Smoothing by bin means:
Bin 1: 9, 9, 9
Bin 2: 22, 22, 22
Bin 3: 29, 29, 29

Smoothing by bin boundaries:
Bin 1: 4, 4, 15
Bin 2: 21, 21, 24
Bin 3: 25, 25, 34

2.Regression:

Data smoothing can also be done by regression, a technique that conforms data values to a

function. Linear regression involves finding the “best” line to fit two attributes (or variables) so that one attribute can be used to predict the other. Multiple linear regression is an extension of linear regression, where more than two attributes are involved and the data are fit to a multidimensionalsurface.

3.Outlier analysis:

Outliers may be detected by clustering, for example, where similar values are organized into groups, or “clusters.” Intuitively, values that fall outside of the set of clusters may be considered outliers

Data Cleaning as a Process

Missing values, noise, and inconsistencies contribute to inaccurate data.

The first step in data cleaning as a process is discrepancy detection

- Discrepancies can be caused by several factors, including poorly designed data entry forms that have many optional fields, human error in data entry, deliberate errors (e.g., respondents not wanting to divulge information about themselves), and data decay (e.g., outdated addresses).
- Discrepancies may also arise from inconsistent data representations and the inconsistent use of codes.
- Errors in instrumentation devices that record data, and system errors, are other sources of discrepancies.
- Errors can also occur when the data are (inadequately) used for purposes other than originally intended.
- There may also be inconsistencies due to data integration (e.g., where a given attribute can have different names in different databases).

Interface College of Computer Applications (ICCA)

Tools for data cleaning

1. Openrefine
2. Trifecta wrangler
3. IBM data refinery
4. Microsoft Excel

Data integration

- Data integration is a critical preprocessing step in data mining that involves combining data from multiple sources into a unified, coherent dataset. It is essentially the process of merging information from diverse sources to create a more comprehensive and valuable view of the data, enabling more sophisticated analysis and insights.

Key Steps in Data Integration

1. **Data Source Identification**: Identify and select the sources from which data will be extracted. These sources can include:
    - Databases (SQL, NoSQL)
    - Data Warehouses
    - APIs
    - Cloud Storage
    - Flat Files (CSV, Excel)
2. **Schema Integration**: This step ensures that the data from different sources are aligned properly by resolving schema-level conflicts such as:
    - **Naming Conflicts**: Different sources might use different names for the same attribute (e.g., "Cust_ID" vs. "CustomerID").
    - **Structural Conflicts**: Different databases may store data in different structures (e.g., hierarchical vs. relational databases).
3. **Data Cleaning**: Since data from different sources may have inconsistencies, errors, and missing values, cleaning is performed to ensure accuracy.
4. **Entity Resolution (Record Linkage)**: This involves identifying and merging duplicate records referring to the same entity.
5. **Data Redundancy Elimination**: Redundant data is removed to improve storage efficiency and reduce inconsistencies.
6. **Data Loading**: After integration, data is loaded into a central repository like a data warehouse or a data lake for further analysis.

Interface College of Computer Applications (ICCA)

Data Transformation

Data transformation involves converting data into a suitable format for analysis. It ensures consistency and standardization across different datasets.

Key Steps in Data Transformation

1. **Data Smoothing**: Techniques such as binning, clustering, and regression are used to remove noise from the data.
2. **Normalization**: This step adjusts the data to fall within a specific range to ensure uniformity. Common normalization techniques include:
    - **Min-Max Normalization**: Scaling data between a specified range (e.g., [0,1] or [-1,1]).
    - **Z-Score Normalization**: Standardizing data using mean and standard deviation.
    - **Decimal Scaling**: Scaling values by dividing by powers of 10.
3. **Aggregation**: Summarizing data by combining values, such as calculating total sales per month instead of per day.
4. **Generalization**: Converting low-level data attributes into higher-level concepts (e.g., changing specific ages into age groups like "Young," "Middle-aged," and "Senior").
5. **Feature Engineering**: Creating new features from existing data to improve the predictive power of machine learning models.
6. **Data Discretization**: Converting continuous data into categorical data by creating bins (e.g., grouping customer ages into ranges like 18-25, 26-35, etc.).
7. **Encoding and Formatting**: Converting categorical data into numerical form using techniques like One-Hot Encoding or Label Encoding.

Importance of Data Integration and Transformation

- **Improves Data Quality**: Removes inconsistencies and ensures a standardized format.
- **Enhances Data Usability**: Makes raw data more meaningful for analysis and modeling.

Interface College of Computer Applications (ICCA)

- **Reduces Redundancy**: Eliminates duplicate records, reducing storage and processing overhead.
- **Prepares Data for Mining Algorithms**: Ensures compatibility with machine learning and statistical models.

DATA REDUCTION

Data reduction techniques can be applied to obtain a reduced representation of the data set that is much smaller in volume, yet closely maintains the integrity of the original data. That is, mining on the reduced data set should be more efficient yet produce the same (or almost the same) analytical results.

Techniques of Data Reduction

1. **Dimensionality Reduction**
2. **Numerosity Reduction**
3. **Data cube Aggregation**
4. **Data compression**
5. **Discretization operation**

1. **Dimensionality Reduction** Transforming data to a lower-dimensional space while preserving important info.

Whenever we encounter weakly important data, we use the attribute required for our analysis. Dimensionality reduction eliminates the attributes from the data set under consideration, thereby reducing the volume of original data. It reduces data size as it eliminates outdated or redundant features. Here are three methods of dimensionality reduction.

i. **Wavelet Transform:** In the wavelet transform, suppose a data vector A is transformed into a numerically different data vector A' such that both A and A' vectors are of the same length. Then how it is useful in reducing data because the data obtained from the wavelet transform can be truncated. The compressed data is obtained by retaining the smallest fragment of the strongest wavelet coefficients. Wavelet transform can be applied to data cubes, sparse data, or skewed data.

ii. **Principal Component Analysis:** Suppose we have a data set to be analyzed that has tuples with n attributes. The principal component analysis identifies k independent tuples with n attributes that can represent the data set. In this way, the original data can be cast on a much smaller space, and dimensionality reduction can be achieved. Principal component analysis can be applied to sparse and skewed data.

iii. **Attribute Subset Selection:** The large data set has many attributes, some of which are irrelevant to data mining or some are redundant. The core attribute subset selection reduces the data volume and dimensionality. The attribute subset selection reduces the volume of data by eliminating redundant and irrelevant attributes. The attribute subset selection ensures that we get a good subset of original attributes even after eliminating the unwanted attributes. The resulting probability of data distribution is as close as possible to the original data distribution using all the attributes.

2. Numerosity Reduction Replacing the actual data with a smaller form

The numerosity reduction reduces the original data volume and represents it in a much smaller form. This technique includes two types parametric and non-parametric numerosity reduction.

i. **Parametric:** Parametric numerosity reduction incorporates storing only data parameters instead of the original data. One method of parametric numerosity reduction is the regression and log-linear method.

- **Regression and Log-Linear:** Linear regression models a relationship between the two attributes by modeling a linear equation to the data set.

i. Log-linear model discovers the relation between two or more discrete attributes in the database. Suppose we have a set of tuples presented in n-dimensional space. Then the log-linear model is used to study the probability of each tuple in a multidimensional space.

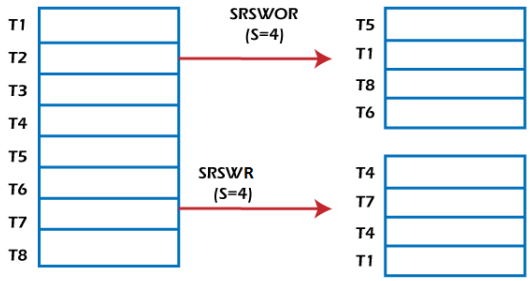
Regression and log-linear methods can be used for sparse data and skewed data.

ii. **Non-Parametric:** A non-parametric numerosity reduction technique does not assume any model. The non-Parametric technique results in a more uniform reduction, irrespective of data size, but it may not achieve a high volume of data reduction like the parametric. There are at least four types of Non-Parametric data reduction techniques, Histogram, Clustering, Sampling, Data Cube Aggregation, and Data Compression.

- **Histogram:** A histogram is a graph that represents frequency distribution which describes how often a value appears in the data. Histogram uses the binning method to represent an attribute's data distribution. It uses a disjoint subset which we call bin or buckets. A histogram can represent a dense, sparse, uniform, or skewed data. Instead of only one attribute, the histogram can be implemented for multiple attributes. It can effectively represent up to five attributes.
- **Clustering:** Clustering techniques groups similar objects from the data so that the objects in a cluster are similar to each other, but they are dissimilar to objects in another cluster. How much similar are the objects inside a cluster can be calculated using a distance function. More is the similarity between the objects in a cluster closer they appear in the cluster. The quality of the cluster depends on the diameter of the cluster, i.e., the max distance between any two objects in the cluster. The cluster representation replaces the original data. This technique is more effective if the present data can be classified into a distinct clustered.
- **Sampling:** One of the methods used for data reduction is sampling, as it can reduce the large data set into a much smaller data sample. Below we will discuss the different methods in which we can sample a large data set D containing N tuples:

a. **Simple random sample without replacement (SRSWOR) of size s:** In this s, some tuples are drawn from N tuples such that in the data set D (s<N). The probability of drawing any tuple from the data set D is 1/N. This means all tuples have an equal probability of getting sampled.

b. **Simple random sample with replacement (SRSWR) of size s:** It is similar to the SRSWOR, but the tuple is drawn from data set D, is recorded, and then replaced into the data set D so that it can be drawn again.



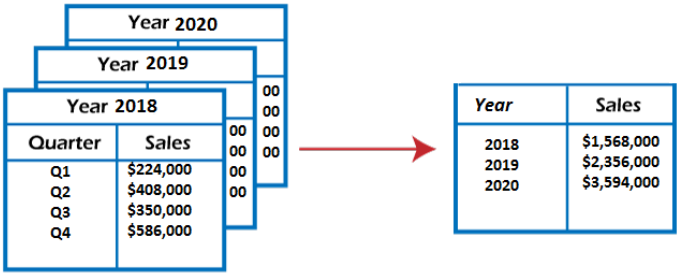
c. **Cluster sample:** The tuples in data set D are clustered into M mutually disjoint subsets. The data reduction can be applied by implementing SRSWOR on these clusters. A simple random sample of size s could be generated from these clusters where s<M.

d. **Stratified sample:** The large data set D is partitioned into mutually disjoint sets called 'strata'. A simple random sample is taken from each stratum to get stratified data. This method is effective for skewed data.

3. Data Cube Aggregation

This technique is used to aggregate data in a simpler form. Data Cube Aggregation is a multidimensional aggregation that uses aggregation at various levels of a data cube to represent the original data set, thus achieving data reduction.

- For example, suppose you have the data of All Electronics sales per quarter for the year 2018 to the year 2022. If you want to get the annual sale per year, you just have to aggregate the sales per quarter for each year. In this way, aggregation provides you with the required data, which is much smaller in size, and thereby we achieve data reduction even without losing any data.

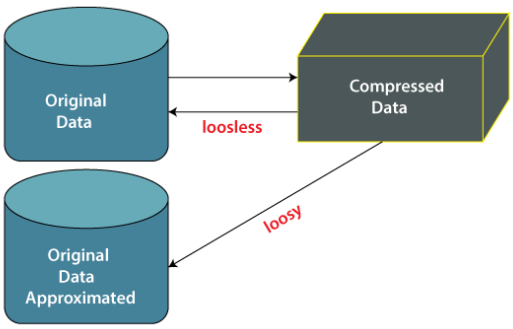


Aggregated Data

The data cube aggregation is a multidimensional aggregation that eases multidimensional analysis. The data cube present precomputed and summarized data which eases the data mining into fast access.

4. Data Compression

Data compression employs modification, encoding, or converting the structure of data in a way that consumes less space. Data compression involves building a compact representation of information by removing redundancy and representing data in binary form. Data that can be restored successfully from its compressed form is called Lossless compression. In contrast, the opposite where it is not possible to restore the original form from the compressed form is Lossy compression. Dimensionality and numerosity reduction method are also used for data compression.



This technique reduces the size of the files using different encoding mechanisms, such as Huffman Encoding and run-length Encoding. We can divide it into two types based on their compression techniques.

i. **Lossless Compression:** Encoding techniques (Run Length Encoding) allow a simple and minimal data size reduction. Lossless data compression uses algorithms to restore the precise original data from the compressed data.

ii. **Lossy Compression: In** lossy-data compression, the decompressed data may differ from the original data but are useful enough to retrieve information from them. For example, the JPEG image format is a lossy compression, but we can find the meaning equivalent to the

original image. Methods such as the Discrete Wavelet transform technique PCA (principal component analysis) are examples of this compression.

Discretization Operation

Convert continuous data into discrete buckets

The data discretization technique is used to divide the attributes of the continuous nature into data with intervals. We replace many constant values of the attributes with labels of small intervals. This means that mining results are shown in a concise and easily understandable way.

i. **Top-down discretization:** If you first consider one or a couple of points (so-called breakpoints or split points) to divide the whole set of attributes and repeat this method up to the end, then the process is known as top-down discretization, also known as splitting.
ii. **Bottom-up discretization:** If you first consider all the constant values as split-points, some are discarded through a combination of the neighborhood values in the interval. That process is called bottom-up discretization.

DATA DISCRETIZATION AND CONCEPT HIERARCHY GENERATION

Data discretization techniques can be used to reduce the number of values for a given continuous attribute by dividing the range of the attribute into intervals.

Data discretization refers to a method of converting a huge number of data values into smaller ones so that the evaluation and management of data become easy. In other words, data discretization is a method of converting attributes values of continuous data into a finite set of intervals with minimum data loss.

Interface College of Computer Applications (ICCA)

1. **Discretization by Binning**

Binning refers to a data smoothing technique that helps to group a huge number of continuous values into smaller values. For data discretization and the development of idea hierarchy, this technique can also be used.

2. **Discretization by Histogram Analysis**

Histogram refers to a plot used to represent the underlying frequency distribution of a continuous data set. Histogram assists the data inspection for data distribution. For example, Outliers, skewness representation, normal distribution representation, etc.

3. **Discretization by Cluster**

clustering can be applied to discretize numeric attributes. It partitions the values into different clusters or groups by following a top-down or bottom-up strategy

4. **Discretization by Decision Tree**

Data discretization refers to a decision tree analysis using a top-down slicing technique. It is done through a supervised procedure. In a numeric attribute discretization, first, you need to select the attribute with the least entropy and then run it with the help of a recursive process. The recursive process divides it into discretized disjoint intervals, from top to bottom, using the same splitting criterion.

5. **Discretization by Correlation Analyses**

Discretizing data by linear regression technique, you can get the best neighboring interval, and then the large intervals are combined to develop a larger overlap to form the final 20 overlapping intervals. It is a supervised procedure.

Concept hierarchy generation

The term hierarchy represents an organizational structure or mapping in which items are ranked according to their levels of importance. In other words, we can say that a hierarchy concept refers to a sequence of mappings with a set of more general concepts to complex concepts.

- It means mapping is done from low-level concepts to high-level concepts.
- For example, in computer science, there are different types of hierarchical systems. A document is placed in a folder in windows at a specific place in the tree structure is the best example of a computer hierarchical tree model. There are two types of hierarchy: top-down mapping and the second one is bottom-up mapping.

Interface College of Computer Applications (ICCA)

Top-down mapping

Top-down mapping generally starts with the top with some general information and ends with the bottom to the specialized information.

Bottom-up mapping

Bottom-up mapping generally starts with the bottom with some specialized information and ends with the top to the generalized information.

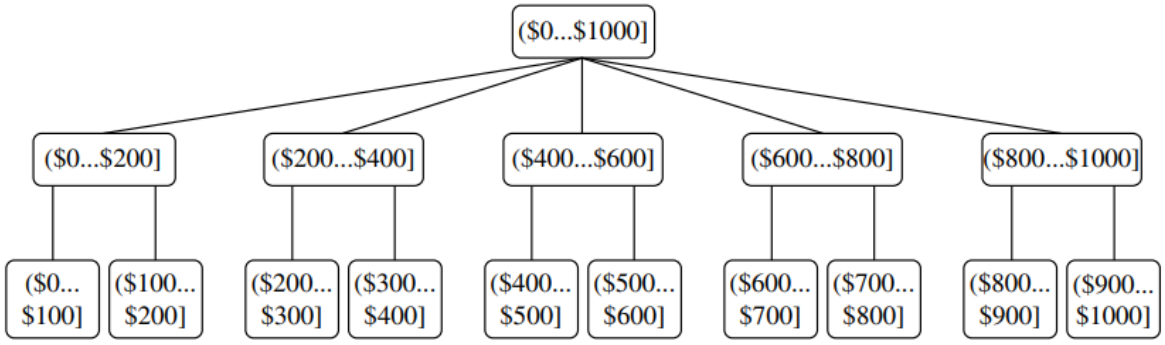


Figure: Concept hierarchy for the attribute price

Types of Concept Hierarchies

1. **Schema Hierarchy:** Schema Hierarchy is a type of concept hierarchy that is used to organize the schema of a database in a logical and meaningful way, grouping similar objects together. A schema hierarchy can be used to organize different types of data, such as tables, attributes, and relationships, in a logical and meaningful way.

2. **Set-Grouping Hierarchy:** Set-Grouping Hierarchy is a type of concept hierarchy that is based on set theory, where each set in the hierarchy is defined in terms of its membership in other sets. Set-grouping hierarchy can be used for data cleaning, data pre-processing and data integration. This type of hierarchy can be used to identify and remove outliers, noise, or inconsistencies from the data and to integrate data from multiple sources.

3. **Operation-Derived Hierarchy:** An Operation-Derived Hierarchy is a type of concept hierarchy that is used to organize data by applying a series of operations or transformations to the data. The operations are applied in a top-down fashion, with each level of the hierarchy representing a more general or abstract view of the data than the level below it. This type of hierarchy is typically used in data mining tasks such as clustering and dimensionality reduction. The operations applied can be mathematical or statistical operations such as aggregation, normalization.

4. **Rule-based Hierarchy:** Rule-based Hierarchy is a type of concept hierarchy that is used to organize data by applying a set of rules or conditions to the data. This type of hierarchy is useful in data mining tasks such as classification, decision-making, and data exploration. It allows to the assignment of a class label or decision to each data

Interface College of Computer Applications (ICCA)

point based on its characteristics and identifies patterns and relationships between different attributes of the data.

Multidimensional Data Model

A Multidimensional Data Model is defined as a model that allows data to be organized and viewed in multiple dimensions, such as time, item, branch, and location, enabling organizations to analyze relationships between different perspectives and entities efficiently.

Data Warehouse Modeling: Data Cube and OLAP

Data warehouses and OLAP tools are based on a **multidimensional data model**. This model views data in the form of a *data cube*.

Data Cube: A Multidimensional Data Model

A **data cube** allows data to be modeled and viewed in multiple dimensions. It is defined by dimensions and facts.In general terms, **dimensions** are the perspectives or entities with respect to which an organization wants to keep records.

For example, *AllElectronics* may create a *sales* data warehouse in order to keep records of the store's sales with respect to the dimensions *time*, *item*, *branch*, and *location*.

These dimensions allow the store to keep track of things like monthly sales of items and the branches and locations at which the items were sold. Each dimension may have a table associated with it, called a **dimension table**, which further describes the dimension.

Interface College of Computer Applications (ICCA)

For example, a dimension table for *item* may contain the attributes *item_name, brand*, and *type*. Dimension tables can be specified by users or experts, or automatically generated and adjusted based on data distributions.

A multidimensional data model is typically organized around a central theme, such as *sales*. This theme is represented by a fact table. **Facts** are numeric measures. Examples of facts for a sales data warehouse include *dollars_sold* (sales amount in dollars), *units_sold* (number of units sold), and *amount_budgeted*.

The **fact table** contains the names of the *facts*, or measures, as well as keys to each of the related dimension tables.

A table or spreadsheet for sales data from *AllElectronics*. In particular, we will look at the *AllElectronics* sales data for items sold per quarter in the city of Vancouver.

These data are shown in Table 4.2. In this 2-D representation, the sales for Vancouver are shown with respect to the *time* dimension (organized in quarters) and the *item* dimension (organized according to the types of items sold). The fact or measure displayed is *dollars_sold* (in thousands).

Table 4.2. 2-D View of Sales Data for *AllElectronics* According to *time* and *item*

location = "Vancouver"				
Empty Cell	***item (type)***			
time (quarter)	***home entertainment***	***computer***	***phone***	***security***
Q1	605	825	14	400
Q2	680	952	31	512
Q3	812	1023	30	501
Q4	927	1038	38	580

Note: The sales are from branches located in the city of Vancouver. The measure displayed is *dollars_sold*) (in thousands).

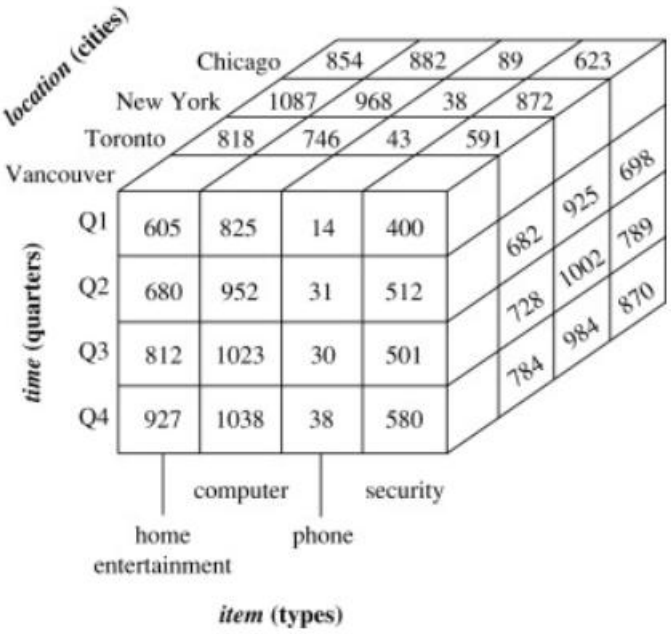
Now, suppose that we would like to view the sales data with a third dimension. For instance, suppose we would like to view the data according to *time* and *item*, as well as *location*, for the cities Chicago, New York, Toronto, and Vancouver. These 3-D data are shown in Table 4.3. The 3-D data in the table are represented as a series of 2-D tables. Conceptually, we may also represent the same data in the form of a 3-D data cube, as in Figure 4.3.also represent the same data in the form of a 3-D data cube, as in Figure 4.3.

Interface College of Computer Applications (ICCA)

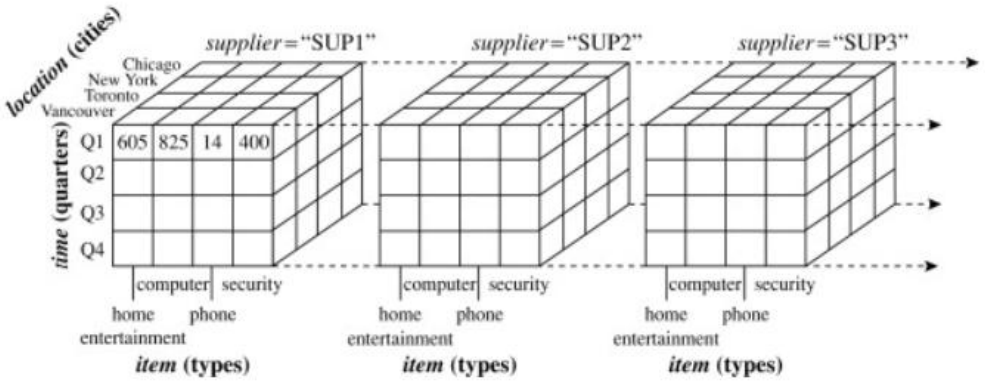
Table 4.3. 3-D View of Sales Data for *AllElectronics* According to *time*, *item*, and *location*

location = "Chicago"					***location = "New York"***				***location = "Toronto"***				***location = "Vancouver"***			
Empty Cell	***item***				***item***				***item***				***item***			
time	***home ent.***	***comp.***	***phone***	***sec.***	***home ent.***	***comp.***	***phone***	***sec.***	***home ent.***	***comp.***	***phone***	***sec.***	***home ent.***	***comp.***	***phone***	***sec.***
Q1	0854	882	89	623	1087	0968	38	0872	818	746	43	591	605	0825	14	400
Q2	0943	890	64	698	1130	1024	41	0925	894	769	52	682	680	0952	31	512
Q3	1032	924	59	789	1034	1048	45	1002	940	795	58	728	812	1023	30	501
Q4	1129	992	63	870	1142	1091	54	0984	978	864	59	784	927	1038	38	580

Note: The measure displayed is *dollars_sold* (in thousands).

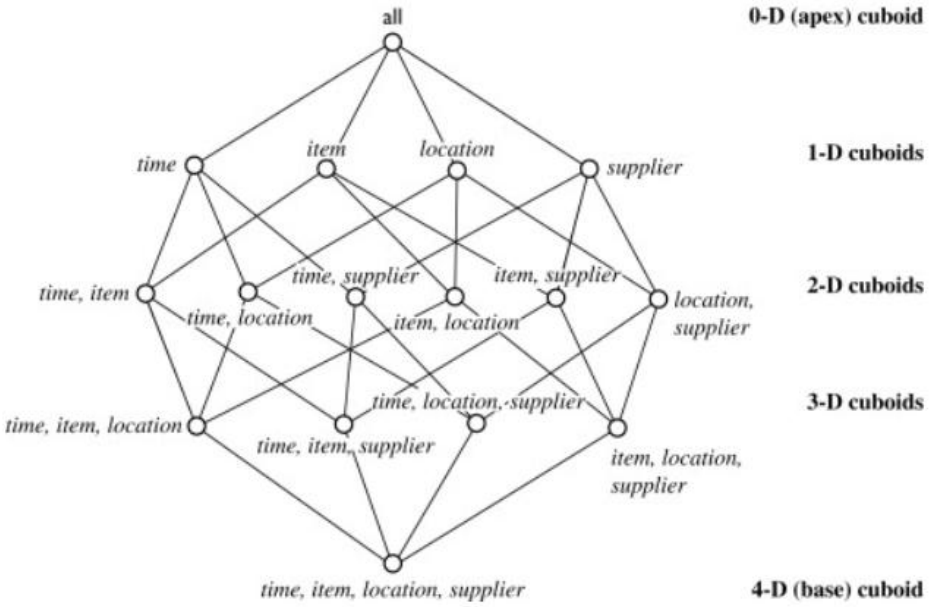


Suppose that we would now like to view our sales data with an additional fourth dimension such as *supplier*. Viewing things in 4-D becomes tricky. However, we can think of a 4-D cube as being a series of 3-D cubes, as shown in Figure 4.4. If we continue in this way, we may display any *n*-dimensional data as a series of (*n* − 1)-dimensional "cubes." The data cube is a metaphor for multidimensional data storage. The actual physical storage of such data may differ from its logical representation. The important thing to remember is that data cubes are *n*-dimensional and do not confine data to 3-D.



Tables 4.2 and 4.3 show the data at different degrees of summarization. In the data warehousing research literature, a data cube like those shown in Figures 4.3 and 4.4 is often referred to as a **cuboid**. Given a set of dimensions, we can generate a cuboid for each of the possible subsets of the given dimensions. The result would form a *lattice* of cuboids, each showing the data at a different level of summarization, or group-by. The lattice of cuboids is

then referred to as a data cube. Figure 4.5 shows a lattice of cuboids forming a data cube for the dimensions *time*, *item*, *location*, and *supplier*.



various multidimensional models are shown: star schema, snowflake schema, and fact constellation.

Star Schema

- A star schema is a type of relational database schema that is composed of a single, central fact table that is surrounded by dimension tables.
- Each dimension in a star schema is represented with only one-dimension table.
- This dimension table contains the set of attributes.
- The following diagram shows the sales data of a company concerning the four dimensions, namely time, item, branch, and location.
- There is a fact table at the center. It contains the keys to each of the four dimensions.
- The fact table also contains the attributes, namely dollars sold and units sold.

Interface College of Computer Applications (ICCA)

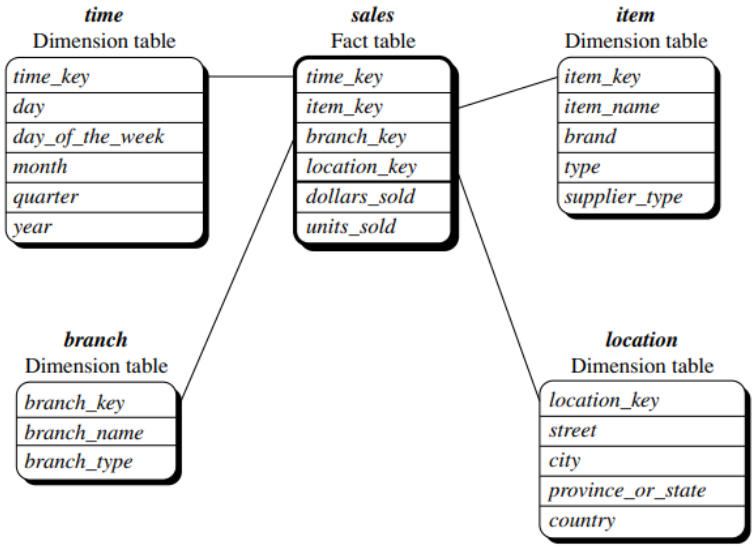


Figure: Star schema of a sales data warehouse

Snowflake Schema

- A snowflake schema is a multi-dimensional data model that is an extension of a star schema, where dimension tables are broken down into subdimensions. Snowflake schemas are commonly used for business intelligence and reporting in OLAP data warehouses, data marts, and relational databases.
- Some dimension tables in the Snowflake schema are normalized.
- The normalization splits up the data into additional tables.
- Unlike Star schema, the dimensions table in a snowflake schema is normalized. For example, the item dimension table in a star schema is normalized and split into two dimension tables, namely item and supplier table.
- Now the item dimension table contains the attributes item_key, item_name, type, brand, and supplier-key.
- The supplier key is linked to the supplier dimension table. The supplier dimension table contains the attributes supplier_key and supplier_type.

Interface College of Computer Applications (ICCA)

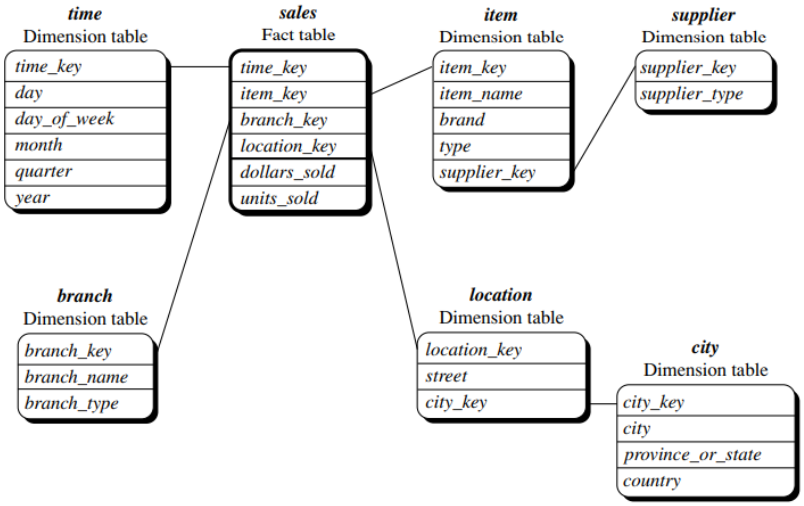


Figure: Snowflake schema of a sales data warehouse

Fact Constellation Schema

- Fact Constellation is a schema for representing multidimensional model. It is a collection of multiple fact tables having some common dimension tables. It can be viewed as a collection of several star schemas
- A fact constellation has multiple fact tables. It is also known as galaxy schema.
- The following diagram shows two fact tables, namely sales and shipping.
- The sales fact table is the same as that in the star schema.
- The shipping fact table has five dimensions, namely item_key, time_key, shipper_key, from_location, to_location.
- The shipping fact table also contains two measures, namely dollars sold and units sold.
- It is also possible to share dimension tables between fact tables. For example, time, item, and location dimension tables are shared between the sales and shipping fact tables.

Interface College of Computer Applications (ICCA)

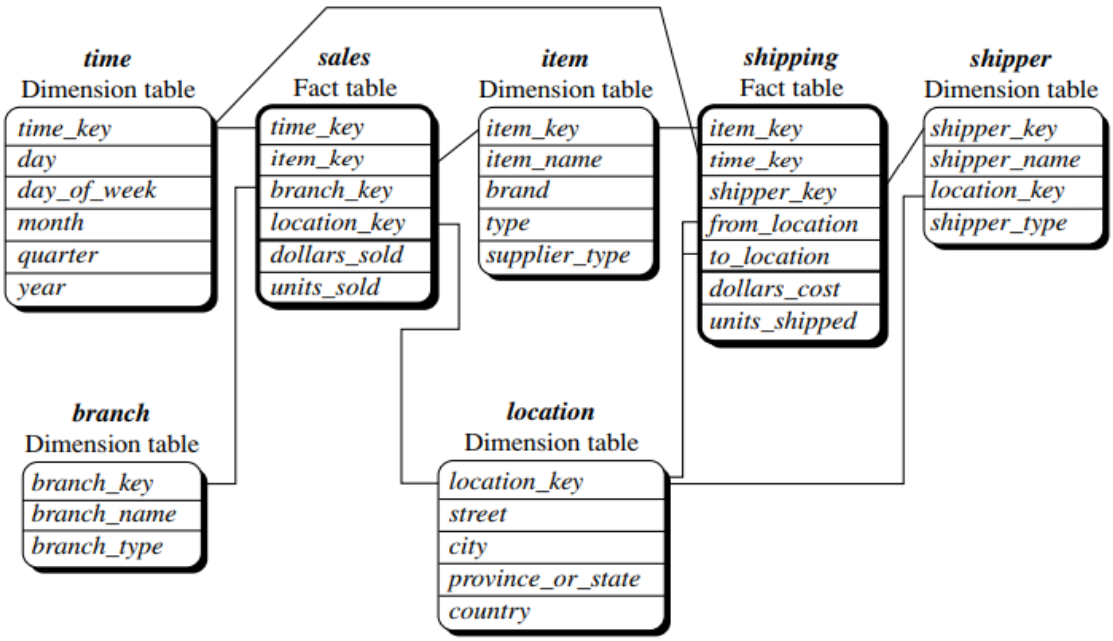


Figure: Fact Constellation schema of a sales shipping data warehouse

ICCA
Interface College of Computer Applications (ICCA)

Unit -3:

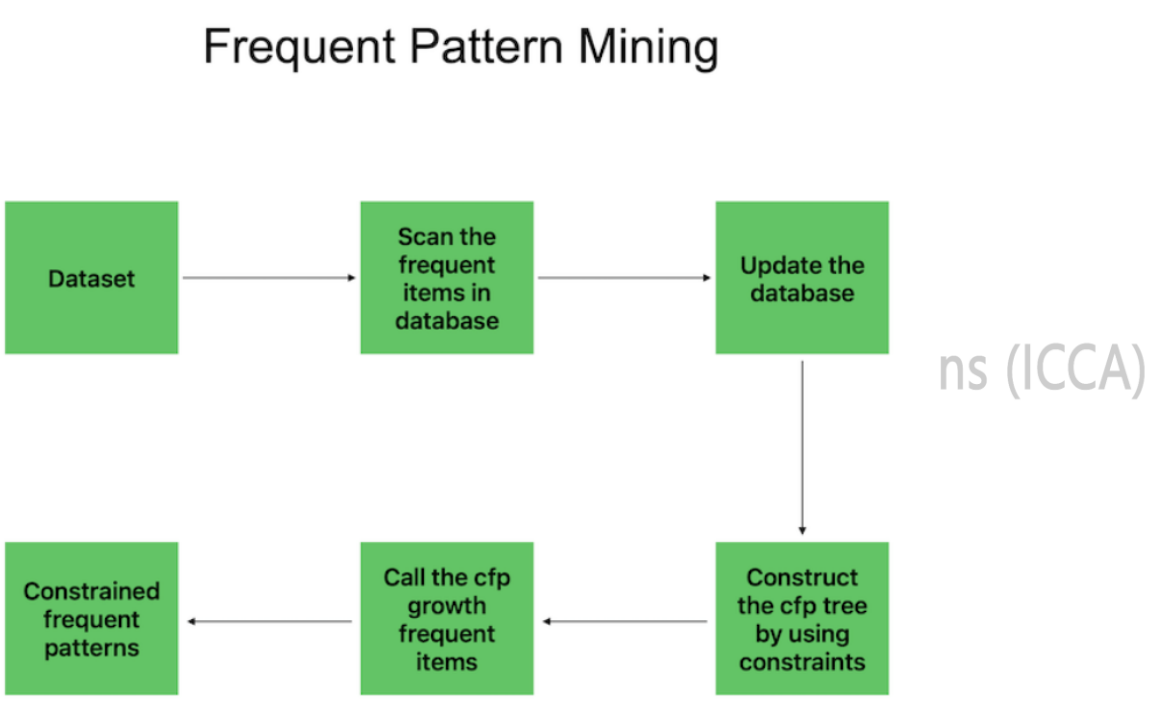
Mining Frequent pattern

Basic Concepts in Frequent Pattern Mining

The technique of frequent pattern mining is built upon a number of fundamental ideas. The analysis is based on transaction databases, which include records or transactions that represent collections of objects. Items inside these transactions are grouped together as itemsets.

The importance of patterns is greatly influenced by support and confidence measurements. Support quantifies how frequently an itemset appears in the database, whereas confidence quantifies how likely it is that a rule generated from the itemset is accurate.

The Apriori algorithm, a popular method for finding recurrent patterns, takes a methodical approach. In order to find no more frequent itemsets, it generates candidate itemsets, prunes the infrequent ones, and then progressively grows the size of the itemsets. The patterns that fulfill the required support criteria are successfully identified through this iterative approach.



Techniques for Frequent Pattern Mining

Apriori Algorithm

One of the most popular methods, the Apriori algorithm, uses a step by step procedure to find frequent item sets. It starts by creating candidate itemsets of length 1, determining their support, and eliminating any that fall below the predetermined cutoff. The method then joins the frequent itemsets from the previous phase to produce bigger itemsets repeatedly.

Once no more common item sets can be located, the procedure is repeated. The Apriori approach is commonly used because of its efficiency and simplicity, but because it requires numerous database scans for big datasets, it can be computationally inefficient.

FP growth Algorithm

A different strategy for frequent pattern mining is provided by the FP growth algorithm. It creates a small data structure known as the FP tree that effectively describes the dataset without creating candidate itemsets. The FP growth algorithm constructs the FP tree recursively and then directly mines frequent item sets from it.

FP growth can be much quicker than Apriori by skipping the construction of candidate itemsets, which lowers the number of runs over the dataset. It is very helpful for sparse and huge datasets.

Eclat Algorithm

Equivalence Class Clustering and bottom up Lattice Traversal are the acronyms for the Eclat algorithm, a well liked frequent pattern mining method. It explores the itemset lattice using a depth first search approach, concentrating on the representation of vertical data formats.

Transaction identifiers (TIDs) are effectively used by Eclat to locate intersections between item sets. This technique is renowned for its ease of use and little memory requirements, making it appropriate for mining frequent itemsets in vertical databases.

Interface College of Computer Applications (ICCA)

Applications of Frequent Pattern Mining

Market Basket Analysis

Market basket analysis frequently mines patterns to comprehend consumer buying patterns. Businesses get knowledge about product associations by recognizing itemsets that commonly appear together in transactions. This knowledge enables companies to improve recommendation systems and crossâ ˆ ’ sell efforts. Retailers can use this program to assist them in making dataâ ˆ ’ driven decisions that will enhance customer happiness and boost sales.

Web usage mining

Web usage mining is examining user navigation patterns to learn more about how people use websites. In order to personalize websites and enhance their performance, frequent pattern mining makes it possible to identify recurrent navigation patterns and session patterns. Businesses can change content, layout, and navigation to improve user experience and boost engagement by studying how consumers interact with a website.

Bioinformatics

The identification of relevant DNA patterns in the field of bioinformatics is made possible by often occurring pattern mining. Researchers can get insights into genetic variants, illness connections, and drug development by examining big genomic databases for recurrent patterns. In order to diagnose diseases, practice personalized medicine, and create innovative therapeutic strategies, frequent pattern mining algorithms help uncover important DNA sequences and patterns.

ASSOCIATION RULE MINING: -

Association rule mining is a data mining technique used to discover interesting relationships or associations between variables in large datasets. These relationships are often expressed as "if-then" statements, known as association rules.

What is Association Rule Mining

Association Rule Mining finds patterns, correlations, or relationships among a set of items in transactional databases, relational databases, or other data repositories.

Interface College of Computer Applications (ICCA)

For example, in a supermarket, ARM can help identify rules like:
"If a customer buys bread, they are likely to buy butter."
"Customers who purchase a laptop often buy a mouse as well."

These rules help businesses optimize marketing strategies, inventory management, and customer engagement.

2. Terminologies in Association Rule Mining

1. Itemset

An **itemset** is a collection of one or more items. For example, in a supermarket dataset, {Milk, Bread, Butter} is an itemset representing a group of products frequently bought together.

- **Singleton Itemset**: An itemset with only one item, e.g., {Milk}.
- **K-itemset**: An itemset containing *k* items, e.g., {Milk, Bread, Butter} is a **3-itemset**.

2. Support

- Support measures the frequency of an itemset in the dataset. It indicates how often an itemset appears in the transaction database.

$$\text{Support (A)} = \frac{\text{Number of transactions containing A}}{\text{Total number of transactions}}$$

For example, if {Milk, Bread} appears in 50 out of 1000 transactions: 0.05

A high support value means that the itemset is frequently occurring in the dataset.

3. Confidence

Confidence measures the reliability of a rule. It is the proportion of transactions that contain itemset **A** which also contain itemset **B** when considering a rule **A** → **B**.

Interface College of Computer Applications (ICCA)

$$\text{Confidence}(A \rightarrow B) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)}$$

For example, if {Milk, Bread} appears in 50 transactions and {Milk} appears in 100 transactions:

A high confidence value indicates a strong correlation between **A** and **B**.

5. Lift

Lift measures how much more likely **B** is to be bought when **A** is bought, compared to when **B** is bought independently.

$$\text{Lift}(A \rightarrow B) = \frac{\text{Support}(A \cup B)}{\text{Support}(A) \times \text{Support}(B)}$$

- **Lift = 1**: No correlation between **A** and **B**.
- **Lift > 1**: Positive correlation (A increases the likelihood of B).
- **Lift < 1**: Negative correlation (A reduces the likelihood of B).

For example, if:

- Support(Milk) = 0.2
- Support(Bread) = 0.3
- Support(Milk, Bread) = 0.1

$$\text{Lift}(Milk \rightarrow Bread) = \frac{0.1}{0.2 \times 0.3} = \frac{0.1}{0.06} = 1.67$$

Since Lift > 1, Milk and Bread are positively correlated.

Apriori Algorithm -

The Apriori algorithm consists of three main phases:

Step 1: Set Minimum Support and Minimum Confidence

- **Minimum Support (min_sup):** Defines how frequently an itemset should appear to be considered significant.
- **Minimum Confidence (min_conf):** Determines the strength of an association rule.

Interface College of Computer Applications (ICCA)

For example:

- min_sup = 50% (An itemset must appear in at least 50% of transactions).
- min_conf = 60% (The rule should have at least 60% confidence).

Step 2: Find Frequent Itemsets Using Iteration

The algorithm follows an iterative approach called **level-wise search**:

Iteration 1: Find Frequent 1-itemsets (L1)

- Count the occurrence of each individual item in the transactions.
- Remove items that do not meet the **min_sup**.

Iteration 2: Generate Candidate 2-itemsets (C2) and Find Frequent 2-itemsets (L2)

- Create **C2 (candidate 2-itemsets)** by combining frequent 1-itemsets.
- Count the occurrences of each candidate itemset in the transactions.
- Remove itemsets that do not meet **min_sup**, forming **L2**.

Iteration 3: Generate Candidate 3-itemsets (C3) and Find Frequent 3-itemsets (L3)

- Use **L2** to generate **C3 (candidate 3-itemsets)**.
- Count the occurrences and filter itemsets below **min_sup** to get **L3**.
- **Repeat** this process until no more frequent itemsets are found.

ICCA
Interface College of Computer Applications (ICCA)

Step 3: Generate Association Rules

Once **frequent itemsets** are found, association rules are generated using **confidence**:

For each frequent itemset **L**, generate rules of the form:

$$A \rightarrow B$$

where:

$$\text{Confidence}(A \rightarrow B) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)}$$

- If **confidence ≥ min_conf**, the rule is considered strong.
- **Lift** can be calculated to measure the rule's significance.

Explain the generation of candidate itemset and frequent pattern with minimum support count = 2 by using Apriori algorithm with below example.

Example

TID	Item Set
10	A, D, E
20	B, C, E
30	A, B, C, E
40	B, E

Step 1: Given Transaction Data

We have a transaction dataset with a **minimum support count = 2**.

	Item Set
10	A, D, E
20	B, C, E
30	A, B, C, E
40	B, E

ICCA
Interface College of Computer Applications (ICCA)

Step 2: Generate Candidate 1-itemsets (C1)

We first count the occurrences of individual items across all transactions.

Item	Count
A	2
B	3
C	2

Item Count

D 1

E 4

Since **minimum support count = 2**, we remove item **D** (as it appears only once).

Frequent 1-itemset (L1):

L1={(A:2),(B:3),(C:2),(E:4)}

Step 3: Generate Candidate 2-itemsets (C2)

We generate all possible **pairs** (combinations of two items) from **L1** and count their occurrences.

TID	Items
10	A, D, E
20	B, C, E
30	A, B, C, E
40	B, E

Now, let's count each **2-itemset**:

Candidate 2-Itemset	Transactions Containing It	Count
(A, B)	{30}	1
(A, C)	{30}	1
(A, E)	{10, 30}	2
(B, C)	{20, 30}	2
(B, E)	{20, 30, 40}	3
(C, E)	{20, 30}	2

Itemset Count

(A, B) 1

(A, C) 1

Itemset	Count
(A, E)	2
(B, C)	2
(B, E)	3
(C, E)	2

Filtering 2-Itemsets with Minimum Support Count

The given **minimum support count** is **2**, so we discard the itemsets that appear less than 2 times.

Itemset	Count	Meets Min Support?
(A, B)	1	(Discarded)
(A, C)	1	(Discarded)
(A, E)	2	(Accepted)
(B, C)	2	(Accepted)
(B, E)	3	(Accepted)
(C, E)	2	(Accepted)

ICCA
Interface College of Computer Applications (ICCA)

So, the **Frequent 2-Itemset (L2)** is:

L2={(A,E:2),(B,C:2),(B,E:3),(C,E:2)}

- We generated **candidate 2-itemsets (C2)** by forming **all possible pairs** from the frequent 1-itemsets (L1).
- We **scanned the database** to count their occurrences.
- We **filtered out** itemsets with **support count < 2**, leaving us with **frequent 2-itemsets (L2)**.
- The frequent 2-itemsets will be used in **Step 4** to generate **candidate 3-itemsets (C3)**.

Step 4: Generating Candidate 3-Itemsets (C3)

- Now that we have **frequent 2-itemsets (L2)**, we move to the next step of the **Apriori Algorithm**: generating **candidate 3-itemsets (C3)** and filtering them based on the minimum support count.
- We generate **3-item combinations** from **L2**.
- From **Step 3**, we obtained the **frequent 2-itemsets (L2):**
- L2={(A,E:2),(B,C:2),(B,E:3),(C,E:2)}

Now, we form **3-item combinations** from these **L2** sets:

- Combine (B, C) and (B, E) → (B, C, E)
- Combine (B, C) and (C, E) → (B, C, E) (Already considered)

Thus, the only **unique candidate 3-itemset (C3)** is:

C3={(B,C,E)}

Counting the Occurrences of Candidate 3-Itemsets (C3)

We check how many times (B, C, E) appears in the transactions.

TID	Item Set	Contains (B, C, E)?
10	A, D, E	(No B, C)
20	B, C, E	(Yes)
30	A, B, C, E	(Yes)
40	B, E	(No C)

Interface College of Computer Applications (ICCA)

Now, we count the occurrences:

Itemset	Count
(B, C, E)	2

Step 5: Generating Candidate 4-Itemsets (C4)

Final Frequent Itemsets

- **L1** (Frequent 1-itemsets): {(A),(B),(C),(E)}
- **L2** (Frequent 2-itemsets): {(A,E),(B,C),(B,E),(C,E)}
- **L3** (Frequent 3-itemsets): {(B,C,E)}

To generate candidate 4 itemset (c4), we need at least two frequent 3 itemset that share a common subset of two items. However in this case L3 contains only one frequent 3

FP Growth in Data Mining

The FP Growth algorithm is a popular method for frequent pattern mining in data mining. It works by constructing a frequent pattern tree (FP-tree) from the input dataset. The FP-tree is a compressed representation of the dataset that captures the frequency and association information of the items in the data.

The algorithm first scans the dataset and maps each transaction to a path in the tree. Items are ordered in each transaction based on their frequency, with the most frequent items appearing first. Once the FP tree is constructed, frequent itemsets can be generated by recursively mining the tree. This is done by starting at the bottom of the tree and working upwards, finding all combinations of itemsets that satisfy the minimum support threshold.

Working on FP Growth Algorithm

The working of the FP Growth algorithm in data mining can be summarized in the following steps:

Interface College of Computer Applications (ICCA)

The working of the FP Growth algorithm in data mining can be summarized in the following steps:

- **Scanthedatabase:**

  In this step, the algorithm scans the input dataset to determine the frequency of each item. This determines the order in which items are added to the FP tree, with the most frequent items added first.
- **Sortitems:**

  In this step, the items in the dataset are sorted in descending order of frequency. The infrequent items that do not meet the minimum support

threshold are removed from the dataset. This helps to reduce the dataset's size and improve the algorithm's efficiency.

- **Construct the FP-tree:**
  In this step, the FP-tree is constructed. The FP-tree is a compact data structure that stores the frequent itemsets and their support counts.
- **Generate frequent itemsets:**
  Once the FP-tree has been constructed, frequent itemsets can be generated by recursively mining the tree. Starting at the bottom of the tree, the algorithm finds all combinations of frequent item sets that satisfy the minimum support threshold.
- **Generate association rules:**
  Once all frequent item sets have been generated, the algorithm post-processes the generated frequent item sets to generate association rules, which can be used to identify interesting relationships between the items in the dataset.

FP Tree

The **FP-tree (Frequent Pattern tree)** is a data structure used in the FP Growth algorithm for frequent pattern mining. It represents the frequent itemsets in the input dataset compactly and efficiently. The FP tree consists of the following components:

- **RootNode:**
  The root node of the FP-tree represents an empty set. It has no associated item but a pointer to the first node of each item in the tree.
- **ItemNode:**
  Each item node in the FP-tree represents a unique item in the dataset. It stores the item name and the frequency count of the item in the dataset.
- **HeaderTable:**
  The header table lists all the unique items in the dataset, along with their frequency count. It is used to track each item's location in the FP tree.
- **ChildNode:**
  Each child node of an item node represents an item that co-occurs with the item the parent node represents in at least one transaction in the dataset.
- **NodeLink:**
  The node-link is a pointer that connects each item in the header table to the

Interface College of Computer Applications (ICCA)

first node of that item in the FP-tree. It is used to traverse the conditional pattern base of each item during the mining process.

The FP tree is constructed by scanning the input dataset and inserting each transaction into the tree one at a time. For each transaction, the items are sorted in descending order of frequency count and then added to the tree in that order. If an item exists in the tree, its frequency count is incremented, and a new path is created from the existing node. If an item does not exist in the tree, a new node is created for that item, and a new path is added to the tree. We will understand in detail how FP-tree is constructed in the next section.

Algorithm by Han

Let's understand with an example how the FP Growth algorithm in data mining can be used to mine frequent itemsets. Suppose we have a dataset of transactions as shown below:

Transaction ID	Items
T1	{M, N, O, E, K, Y}
T2	{D, O, E, N, Y, K}
T3	{K, A, M, E}
T4	{M, C, U, Y, K}
T5	{C, O, K, E, I}

Let's scan the above database and compute the frequency of each item as shown in the below table.

Interface College of Computer Applications (ICCA)

Item	Frequency
A	1
C	2
D	1
E	4
I	1
K	5
M	3
N	2
O	3

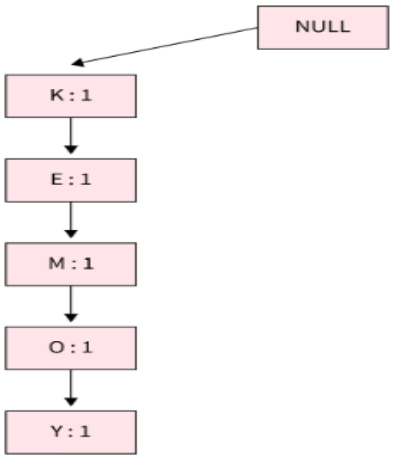
U	1
Y	3

Let's consider minimum support as 3. After removing all the items below minimum support in the above table, we would remain with these items - {K: 5, E: 4, M : 3, O : 3, Y : 3}. Let's re-order the transaction database based on the items above minimum support. In this step, in each transaction, we will remove infrequent items and re-order them in the descending order of their frequency, as shown in the table below.

Transaction ID	Items	Ordered Itemset
T1	{M, N, O, E, K, Y}	{K, E, M, O, Y}
T2	{D, O, E, N, Y, K}	{K, E, O, Y}
T3	{K, A, M, E}	{K, E, M}
T4	{M, C, U, Y, K}	{K, M, Y}
T5	{C, O, K, E, I}	{K, E, O}

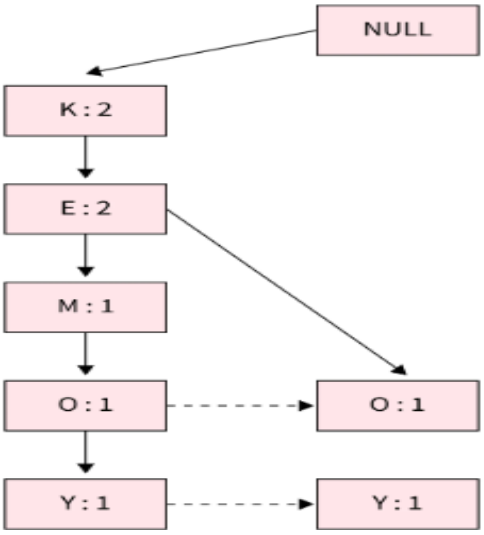
Now we will use the ordered itemset in each transaction to build the FP tree. Each transaction will be inserted individually to build the FP tree, as shown below -

- **First Transaction {K, E, M, O, Y}:** In this transaction, all items are simply linked, and their support count is initialized as 1.

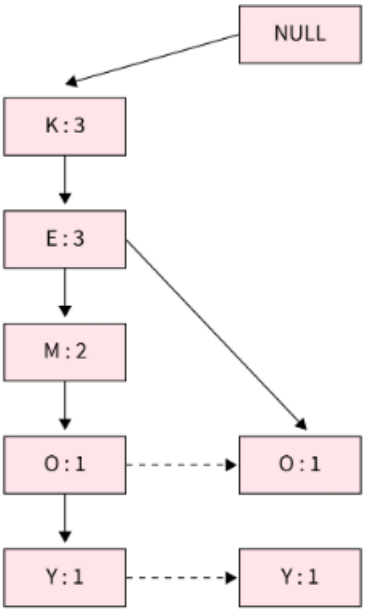


- **Second Transaction {K, E, O, Y}:** In this transaction, we will increase the support count of K and E in the tree to 2. As **no direct link** is available from E to O, we will insert a new path for O and Y and initialize their support count as 1.

mputer Applications (ICCA)

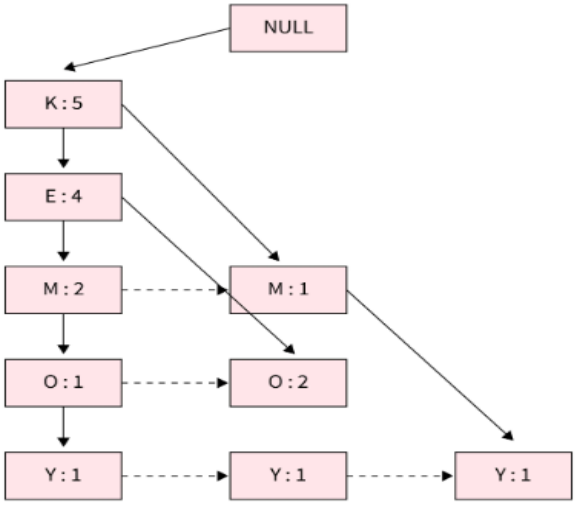


- **Third Transaction {K, E, M}:** After inserting this transaction, the tree will look as shown below. We will increase the support count for K and E to 3 and for M to 2.



Int CA nputer Applications (ICCA)

- **Fourth Transaction {K, M, Y}** and **Fifth Transaction {K, E, O}:** After inserting the last two transactions, the FP-tree will look like as shown below:



Now we will create a **Conditional Pattern Base** for all the items. The conditional pattern base is the path in the tree ending at the given frequent item. For example, for item O, the paths {K, E, M} and {K, E} will result in item O. The conditional pattern base for all items will look like as shown below table:

Item	Conditional Pattern Base
Y	{K, E, M, O : 1}, {K, E, O : 1}, {K, M : 1}
O	{K, E, M : 1}, {K, E : 2}
M	{K, E : 2}, {K : 1}
E	{K : 4}
K	

Now for each item, we will build a conditional frequent pattern tree. It is computed by identifying the set of elements common in all the paths in the conditional pattern base of a given frequent item and computing its support count by summing the support counts of all the paths in the conditional pattern base. The conditional frequent pattern tree will look like this as shown below table:

Item	Conditional Pattern Base	Conditional FP Tree
Y	{K, E, M, O : 1}, {K, E, O : 1}, {K, M : 1}	{K : 3}
O	{K, E, M : 1}, {K, E : 2}	{K, E : 3}
M	{K, E : 2}, {K: 1}	{K : 3}
E	{K: 4}	{K: 4}
K		

Interface College of Computer Applications (ICCA)

From the above conditional FP tree, we will generate the frequent itemsets as shown in the below table:

Item	Frequent Patterns
Y	{K, Y - 3}
O	{K, O - 3}, {E, O - 3}, {K, E, O - 3}
M	{K, M - 3}
E	{K, E - 4}

Mining Frequent Itemsets using the Vertical Data Format (Eclat)

Eclat (Equivalence Class Transformation) is a depth-first search algorithm for discovering frequent itemsets in a transaction database. Unlike the Apriori algorithm which uses a horizontal data format, Eclat uses a vertical data format, which makes it significantly more efficient for many datasets.

The Vertical Data Format

In traditional horizontal data format, data is represented as:

- Transaction ID → List of items

In vertical data format used by Eclat:

Interface College of Computer Applications (ICCA)

- Item → List of transaction IDs (TID-list)

This vertical representation allows for much faster computation of support counts through simple set intersections.

How Eclat Works

1. Initial Transformation: Convert the transaction database from horizontal to vertical format
2. Support Counting: Compute the support of an itemset by intersecting the TID-lists

3. Candidate Generation: Generate candidate itemsets using a depth-first approach
4. Pruning: Eliminate candidates that don't meet the minimum support threshold

Algorithm Steps in Detail

1. Convert to Vertical Format

For each item in the database, create a list of transaction IDs where the item appears.

Example:

Horizontal format:

T1: {A, B, C}

T2: {A, C}

T3: {A, D}

T4: {B, C, D}

Vertical format:

Interface College of Computer Applications (ICCA)

A: {T1, T2, T3}

B: {T1, T4}

C: {T1, T2, T4}

D: {T3, T4}

2. Find Frequent 1-itemsets

Count the length of each TID-list and keep only items that meet the minimum support threshold.

3. Generate Frequent k-itemsets (k ≥ 2)

For each pair of frequent (k-1)-itemsets that share a common prefix of length (k-2):

- The new k-itemset is formed by joining the two (k-1)-itemsets
- The TID-list of the new itemset is the intersection of the TID-lists of the two (k-1)-itemsets
- The support of the new itemset is the length of its TID-list

4. Recursively Continue

Repeat step 3 in a depth-first manner until no more frequent itemsets can be generated.

Example Walkthrough

Let's use the example database from above with minimum support = 2:

Step 1: Convert to vertical format

A: {T1, T2, T3} (support = 3)

B: {T1, T4} (support = 2)

C: {T1, T2, T4} (support = 3)

D: {T3, T4} (support = 2)

Interface College of Computer Applications (ICCA)

All 1-itemsets are frequent (support ≥ 2).

Step 2: Generate frequent 2-itemsets

AB: {T1} = {T1, T2, T3} ∩ {T1, T4} (support = 1) - Not frequent

AC: {T1, T2} = {T1, T2, T3} ∩ {T1, T2, T4} (support = 2) - Frequent

AD: {T3} = {T1, T2, T3} ∩ {T3, T4} (support = 1) - Not frequent

BC: {T1, T4} = {T1, T4} ∩ {T1, T2, T4} (support = 2) - Frequent

BD: {T4} = {T1, T4} ∩ {T3, T4} (support = 1) - Not frequent

CD: {T4} = {T1, T2, T4} ∩ {T3, T4} (support = 1) - Not frequent

Step 3: Generate frequent 3-itemsets Since we have only two frequent 2-itemsets (AC and BC) and they don't share a common prefix, we can't generate any 3-itemsets.

Result: Frequent itemsets

A (support = 3)

B (support = 2)

C (support = 3)

D (support = 2)

AC (support = 2)

BC (support = 2)

Advantages of Eclat

1. Efficiency: Significantly faster than Apriori for many datasets due to:
    - Avoiding multiple database scans
    - Computing support through simple set intersections
    - Depth-first search approach reducing memory requirements
2. Memory Usage: The vertical data format is often more compact, especially for sparse datasets
3. Simplicity: The algorithm is conceptually simpler than many other frequent itemset mining algorithms

Interface College of Computer Applications (ICCA)

Variations and Optimizations

1. dEclat: Uses diffsets (difference of TID-lists) instead of intersections to further improve performance
2. Parallel Eclat: Distributes the computation across multiple processors

3. Memory optimization: Various techniques to reduce the memory footprint when dealing with very large datasets

MINING VARIOUS KINDS OF ASSOCIATION RULES

Mining Multilevel Association Rules

- For many applications, it is difficult to find strong associations among data items at low or primitive levels of abstraction due to the sparsity of data at those levels.
- Strong associations discovered at high levels of abstraction may represent commonsense knowledge.
- Moreover, what may represent common sense to one user may seem novel to another.
- Therefore, data mining systems should provide capabilities for mining association rules at multiple levels of abstraction, with sufficient flexibility for easy traversal among different abstraction spaces.
- Using uniform support for all levels.
- Using reduced minimum support at lower levels.
- Using item group-based minimum support.

Mining Multidimensional Association Rules from Relational Databases and Data Warehouses

Mining multidimensional association rules from relational databases and data warehouses involves discovering associations among multiple dimensions or attributes in a dataset that is structured in a relational or multidimensional database format.

Interface College of Computer Applications (ICCA)

Mining Multidimensional Association Rules Using Static Discretization of Quantitative Attributes

Mining multidimensional association rules using static discretization of quantitative attributes involves transforming continuous numerical data into discrete intervals and then discovering association rules among these discretized attributes across multiple dimensions.

- The intervals are created statically according to the user.

Mining Quantitative Association Rules

Mining quantitative association rules involves discovering interesting relationships between quantitative variables in a dataset.

- The intervals ae created dynamically means generated by the algorithms by using binning methods.

ASSOCIATION MINING TO CORRELATION ANALYSIS

- As we have seen so far, the support and confidence measures are insufficient at filtering out uninteresting association rules.
- To tackle this weakness, a correlation measure can be used to augment the support–confidence framework for association rules.
- This leads to correlation rules of the form

  A ⇒ B [support, confidence, correlation].

- That is, a correlation rule is measured not only by its support and confidence but also by the correlation between itemsets A and B.

Lift is a simple correlation measure that is given as follows.

- The occurrence of itemset A is independent of the occurrence of itemset B if P(A ∪B) = P(A)P(B); otherwise, itemsets A and B are dependent and correlated as events.
- This definition can easily be extended to more than two item sets.
- The lift between the occurrence of A and B can be measured by computing

$$lift(A, B) = \frac{P(A \cup B)}{P(A)P(B)}.$$

Interface College of Computer Applications (ICCA)

- If the resulting value of equation is less than 1, then the occurrence of A is negatively correlated with the occurrence of B, meaning that the occurrence of one likely leads to the absence of the other one.
- If the resulting value is greater than 1, then A and B are positively correlated, meaning that the occurrence of one implies the occurrence of the other.
- If the resulting value is equal to 1, then A and B are independent and have no correlation.

CONSTRAINT-BASED ASSOCIATION MINING.

A data mining process may uncover thousands of rules from a given set of data, most of which end up being unrelated or uninteresting to the users. Often, users have a good sense of which "direction" of mining may lead to interesting patterns and the "form" of the patterns or rules they would like to find. Thus, a good heuristic is to have the users specify such intuition or expectations as constraints to confine the search space. This strategy is known as constraint-based mining.

The constraints can include the following:

- **Knowledge type constraints:** These specify the type of knowledge to be mined, such as association, regression or correlation.

- **Data constraints:** Specifies the type of data on which you want to generate the rules. These specify the set of task-relevant data.

- **Dimension/level constraints:** These specify the desired dimensions (or attributes) of the data, or levels of the concept hierarchies, to be used in mining.

- **Interestingness constraints:** These specify thresholds on statistical measures of rule interestingness, such as support, confidence, and correlation.

- **Rule constraints:** These specify the form of rules to be mined. Such constraints may be expressed as metarules (rule templates), as the maximum or minimum number of predicates that can occur in the rule antecedent or consequent, or as relationships among attributes, attribute values, and/or aggregates.

Interface College of Computer Applications (ICCA)

Interface College of Computer Applications (ICCA)